The Story of the Himalayan Glaciers

Aparna Sharma
aparna.ramgopalsharma@students.iiserpune.ac.in

Hello everyone!
I am writing this document with some basic explanations and interesting facts about glaciers, in particular Himalayan glaciers.
The questions we will try to answer through the exhibit are the ones I already discussed with you:
What are glaciers?
How does a glacier work? Or What are the processes happening in a glacier?
What will be the future of glaciers like?

Since you people did not ask me many questions, I am writing this in the form of a play to make things interesting.
Sorry if I got any of your names wrong.

What are glaciers?

Glaciers are huge bodies of ice that flow due to gravity.
I am not explaining this further.
For this question, let's do an activity. You can all search, make a chart or some drawing. We will keep that there, and you can use that to explain what glaciers are.

Aparna: Now that you all know what glaciers are, let's learn more by asking some questions.

Bhavana: How does a glacier form?

Aparna: That's an important question.
In the cold places in our case high up in a mountain like Mount Everest, when winter comes, snowfall happens. Then when the summer comes, the snow in the lower part of mountains melts but the snow in upper part does not. Again winter comes, and more snowfall happens. If this continues a lot of snow will be on the upper part of a mountain. But gravity comes into action. Rivers from the top of a mountain will flow down due to gravity similarly ice flows down due to gravity. But it flows very slowly so you can't see it flowing. And thus a glacier forms as a moving body of ice. The end part of a glacier is called the terminus of the glacier as the glacier terminates here. In this picture below you can see the terminus of Khumbu glacier at Mount Everest.
A new keyword to remember here is **glacier terminus.**


Khumbu glacier, Everest base camp

Source: SwissEduc

Here is a link with different glacier names and images in India
[List of glaciers in India - Wikipedia]

Jai: What do scientists study about in a glacier?

Aparna: That is a completely valid question.
Glaciers are beautiful landscapes as you can see in the above picture. But if we want to systematically study them we need to define a system. Let’s consider an example of your bank account. When you get your salary(in future), your bank account’s balance increases, and you use that money the entire month. When the month ends, you will get your salary again. In this banking system, there is an input of salary and output when you spend your money.
Similarly, in a glacier system, there is an input and output. Input is the snowfall in winter, output is the melting in summer.

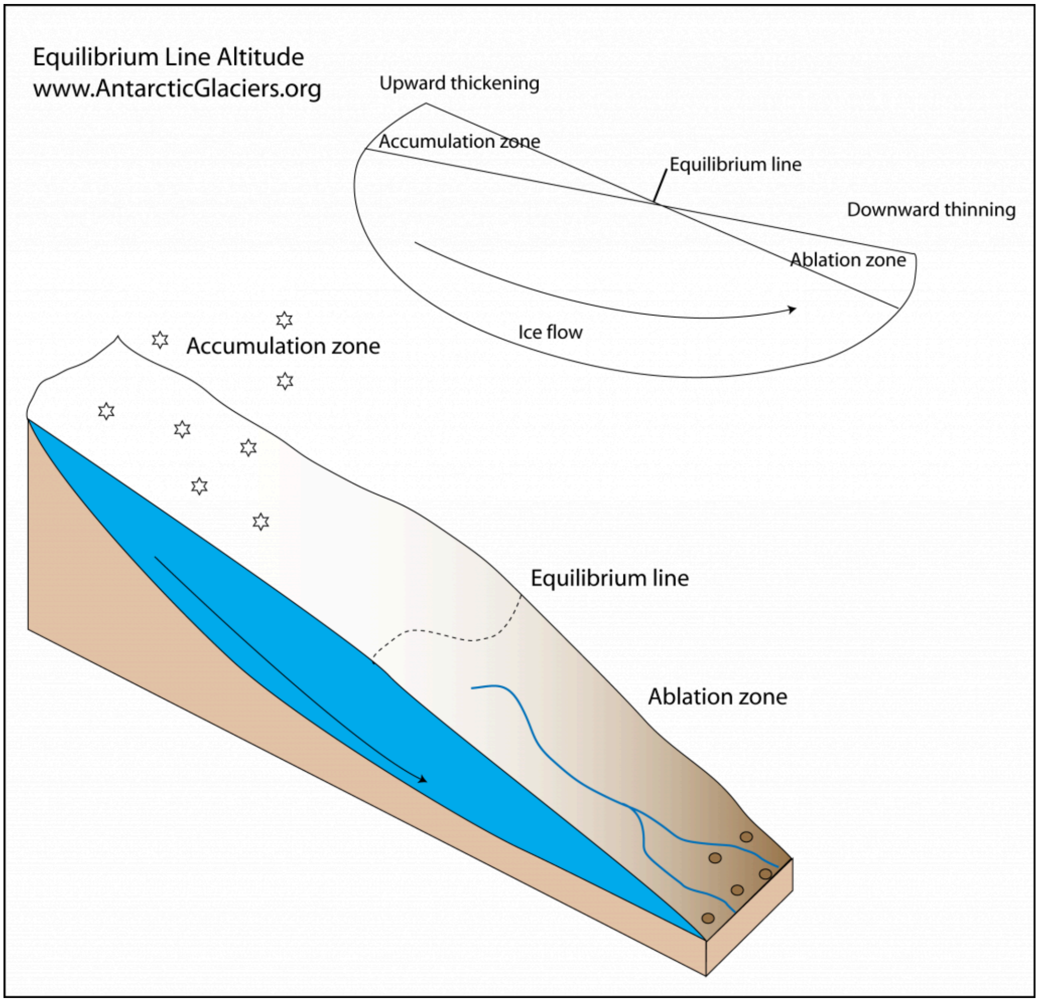
As you go down from a mountain temperature increases and thus melting of ice also increases. In the upper part of the glacier the melting is so less due to low temperature compared to snowfall. You can look at the upper part as you are getting more salary and spending less so your money in the bank will increase. That part of a glacier is called the accumulation zone.

The lower part where there is more melting than snowfall is called the ablation zone that is here you spend more so you won’t have any money left in your bank account. In the middle there is a place snowfall=melting that place is in equilibrium there whatever you earn the same amount you spend- whatever snowfall happens the same amount is melting. Thus we study glaciers by dividing them into these different zones.

In this way, a glacier attains a balance whatever melting is happening in the ablation zone is compensated by the ice flowing from the accumulation zone. Think of it as you spent all your money you get some money from you parents who have more in their savings and thus you still can sustain your lifestyle.

The keywords to remember here are **accumulation zone, ablation zone, equilibrium line.**

Observe the image below at the top the snowfall depicts that there is more snowfall than melting and at the bottom the blue lines indicate melted water showing that there is more melt than snowfall.



Divya: Then why are people talking about climate change and glaciers? Does climate change affect the glaciers?

Aparna: Yes climate change affects the glaciers by disrupting this balance. Due to rising temperature more melting happens. The glacier reduces in size. We call it glacier retreat. But climate change is not the only factor affecting glacier retreat some glaciers retreat faster some retreat slower. It depends on the climatic condition the amount of snowfall, the mountain slope, how high the mountain is. So there were three major things- snowfall to add snow to glacier,

melting that melts the ice and the ice flow all of these together tells us what will happen to a glacier in the future.
The keyword here is **glacier retreat**

Ayush: But why should we be concerned with the retreating glaciers?

Aparna: Retreating glaciers are of very much relevant to us. Three major threats due to retreating glaciers are

1. Sea level rise that can submerge coastal areas. So the ice is melting more, that means there will be more water where will that water go? It will flow into the river. But where does the water in all the rivers go? It goes into the sea and thus sea level will rise.
2. Water security- many people in the himalayan regions depend on glaciers as a source of water. During summer when there is no rainfall the snow and ice melt water can act as a source of water in many places. There are hydropower projects constructed on mountains that utilize the glacier water to generate electricity. As the glacier melts more and more water will increase first but there will come a time when the glaciers will disappear then from where will those places get water in the summer with no rainfall.
3. The final threat is of glacial lake outburst flood. More about this in the video I shared! https://youtu.be/BDPbtP-0AW8?si=Il4FsfGg0WdsPrZl

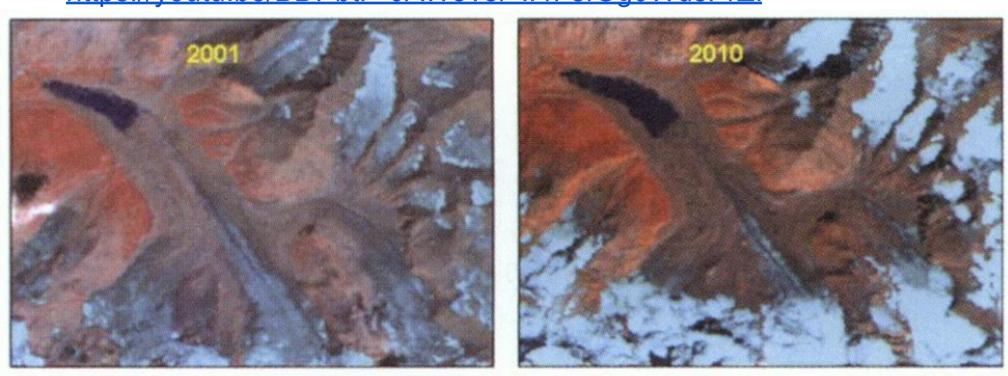


Figure 5. Snout of a glacier in Bhaga basin (Himachal region) showing retreat during 2001–2010.

Source: Bahuguna et al 2014, Current Science
Retreating glaciers and growing lake- Notice the area occupied by glaciers
Don’t worry if you are not able to see the difference yet I will explain the images to you on the science day also.

The keywork here is **GLOF.**
Don’t worry if you don’t understand things. This is all very new to you. I think reading this much is sufficient. If you want you can google search the keywords I mentioned. See you on the National Science Day!